

Solving Problems with EquationsWorksheet 7A.01.05 **ANSWER KEY**

For questions 1 and 2, rewrite the expression by applying the distributive property.

1. $6(x + 4)$

$$6x + 24$$

6 multiplies both terms inside the parentheses.

2. $(b - 9)(-4)$

$$-4b + 36$$

$(-4)(b) = -4b$ and $(-4)(-9) = +36$. The factor on the right multiplies both terms just the same.

Equations of the form $a(b + c) = d$ can be solved two ways: distribute the a first, or divide both sides by a first.

3. Solve $4(1 - 3x) = 52$ using each method.

Method 1 Distribute first	Method 2 Divide both sides first
$4 - 12x = 52$ $-12x = 48$ $x = -4$	$1 - 3x = 13$ $-3x = 12$ $x = -4$

Both methods give $x = -4$. Check: $4(1 - 3(-4)) = 4(1 + 12) = 4(13) = 52$.

For questions 4 through 7, solve using your preferred method, then check your solution by substitution.

4. $\frac{3}{10}(x + 2) = 12$

$$x + 2 = 12 \div (3/10) = 12 \times (10/3) = 40$$

$$x = 38$$

$$\text{Check: } (3/10)(38 + 2) = (3/10)(40) = 12 \checkmark$$

Dividing first is quicker here. Distributing gives $(3/10)x + 3/5 = 12$, which works but is messier.

5. $0.2(r - 3) = -10$

$$r - 3 = -10 \div 0.2 = -50$$

$$r = -47$$

$$\text{Check: } 0.2(-47 - 3) = 0.2(-50) = -10 \checkmark$$

Dividing by 0.2 is the same as multiplying by 5.

6. $\frac{3}{4} (x + 5\frac{1}{2}) = \frac{5}{8}$

$$x + 5\frac{1}{2} = (5/8) \div (3/4) = (5/8)(4/3) = 5/6$$

$$x = 5/6 - 11/2 = 5/6 - 33/6 = -28/6 = -4\frac{2}{3}$$

$$\text{Check: } (3/4)(-4\frac{2}{3} + 5\frac{1}{2}) = (3/4)(5/6) = 5/8 \checkmark$$

Multiplying by the reciprocal $4/3$ is the efficient move. Answer as a mixed number: $-4\frac{2}{3}$.

7. $911.32 = 21.74p + 780.88$

$$911.32 - 780.88 = 21.74p$$

$$130.44 = 21.74p$$

$$p = 6$$

$$\text{Check: } 21.74(6) + 780.88 = 130.44 + 780.88 = 911.32 \checkmark$$

No parentheses here — this one is a plain two-step equation, and the numbers are only large, not hard.

A local plumber charges a \$75 fee for a home visit plus \$66.24 per hour while working. The total bill was \$339.96.

8. Write an equation for this situation. Use h for the number of hours the plumber worked.

$$75 + 66.24h = 339.96$$

The \$75 is charged once, so it is the constant. The \$66.24 depends on h , so it is the coefficient.

9. Solve the equation.

$$66.24h = 264.96$$

$$h = 4$$

$$\text{Check: } 75 + 66.24(4) = 75 + 264.96 = 339.96 \checkmark$$

10. Explain what the solution means in this situation.

The plumber worked for 4 hours.

Watch for students who answer “\$264.96.” That is what the hours cost, not the number of hours.